U.G.C. Major Research Project Summary Report

"Design of A New Algorithm to Preserve the Privacy in Data Publishing Against Record and Attribute linkage attacks"

(F.No. 41-637/2012(SR) dt. 16-7-2012)

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{A State University Graded Category-I by MHRD-UGC & A+ Grade by NAAC (CGPA: 3.64)}

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Objectives:

The prime objective of this project is to preserve individuals' privacy in data publishing with minimal information loss. Anonymization techniques are applied over raw datasets to transform them into anonymized dataset so that their privacy is preserved. Data utilization should not be degraded while preserving the privacy. Data utlilization is a major concern for researchers while applying anonymization techniques. The very purpose of data publishing is to make it available for researchers. This project has focused on increasing data utilization without compromosing the privacy of individuals.

Methods Designed:

Three new methods have been designed to meet the objectives "Improving Privacy Gain" and "Reducing Information Loss".

The first method proposes a new anonymization technique that improves privacy and reduces information loss through record elimination process compared to existing K-anonymity technique.

The second method is proposed for incremental data publishing scenarios. It improves information loss through local recoding and grouping of sensitive data items. This method reduces the information loss very much compared to existing methods for incremental data sets.

The third method is proposed to generalize the Classification Data model to achieve Anonymity for Data Privacy.

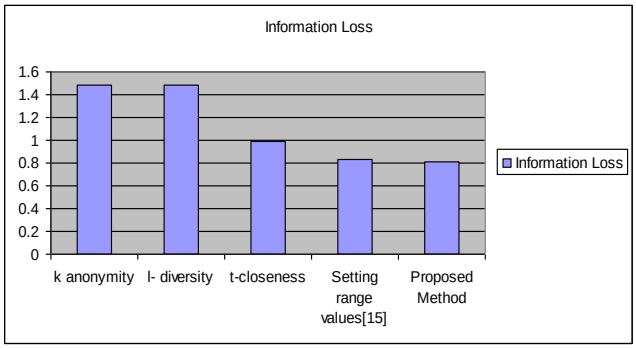
1. Outcome of the Research Paper "Anonymization technique through record elimination to preserve privacy of published data" PRIME 2013, Periyar University, Salem, Tamilnadu conference proceedings published in **IEEE Xplore Digital Library**, Feb 2013

Anonymyzed Medical DataSet

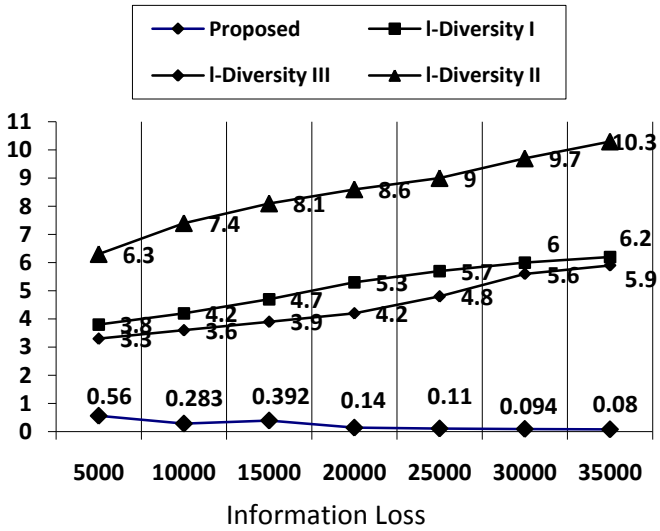
Sno	Zipcode	Age	Salary	Disease	
1	4760*	28<=30	4000	Gastritis	- C_1
2	4760*	28<=32	7000	Bronchitis	
3	4760*	30<=32	10000	Bronchitis	
4	4767*	29<=36	5000	Gastric ulcer	- C_2
6	4767*	29<=36	9000	Pneumonia	
7	4790*	31<=36	6000	Gastritis	- C_3
8	4790*	31<=52	8000	Bronchitis	
9	4790*	36<=52	11000	Flu	

Methods	Information Loss	Privacy Gain	ILPG
k anonymity	1.488	12	.124
l- diversity	1.488	10.5	.1417
t-closeness	.990	10.5	.0942
Setting range values[15]	.832	11.7	.0711
Proposed Method	**.809**	**10.8**	**.0749**

Comparison of Information loss metric between proposed method and existing methods



2. Outcome of the Research Paper published in "Anonymization and Reducing Information loss in Incremental Dataset through Grouping and Local Recoding" published in Australian Journal of Basic and Applied Sciences, **Australian Journal of Basic and Applied Sciences**, 8(4) April 2013, Pages: x-x.



3. Outcome of the Research paper "Generalize the Classification Data model to achieve Anonymity for Data Privacy", Presented in **International Conference** on Computer Engineering and Mathematical Science (ICCEMS 2013), **Kualalumpur, Malaysia**, December 2013.

Raw dataset

No of Recs	Age	Gender	Blood Pressure	Problem
5	29	Male	Yes	Yes
5	28	Male	No	No
4	29	Female	Yes	Yes
1	36	Male	Yes	Yes
1	52	Female	No	No
1	36	Female	No	Yes
3	30	Male	Yes	No
1	36	Male	No	No
1	32	Female	Yes	Yes

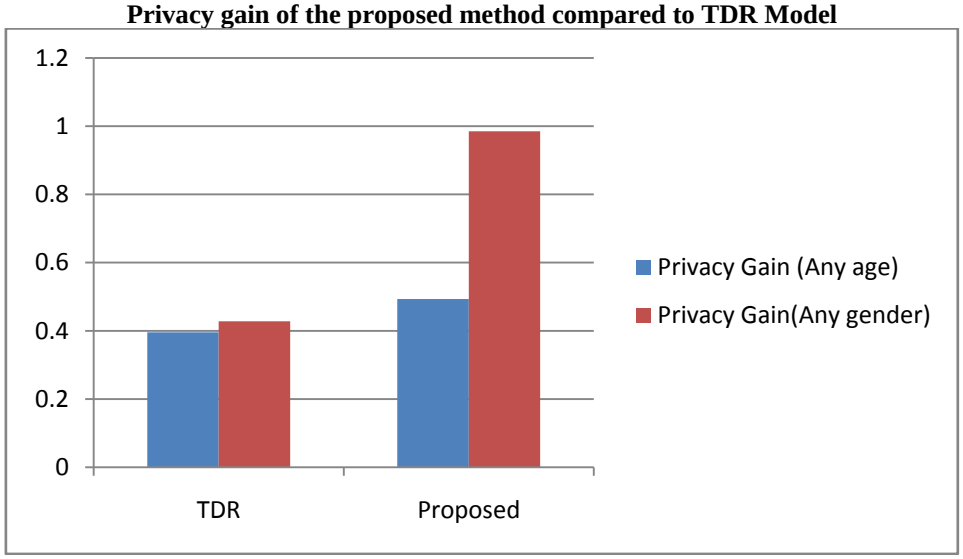
Anonymized Dataset T^*

Age	Gender	Blood Pressure
28<=29	5M 4F	9Y 5N
28<=30	5M 4F	9Y 5N
29<=32	5M 4F	9Y5N
30<=36	3M 1F	3Y1N
32<=52	2M1F	1Y 2N
36<=52	2M 1F	2Y1N

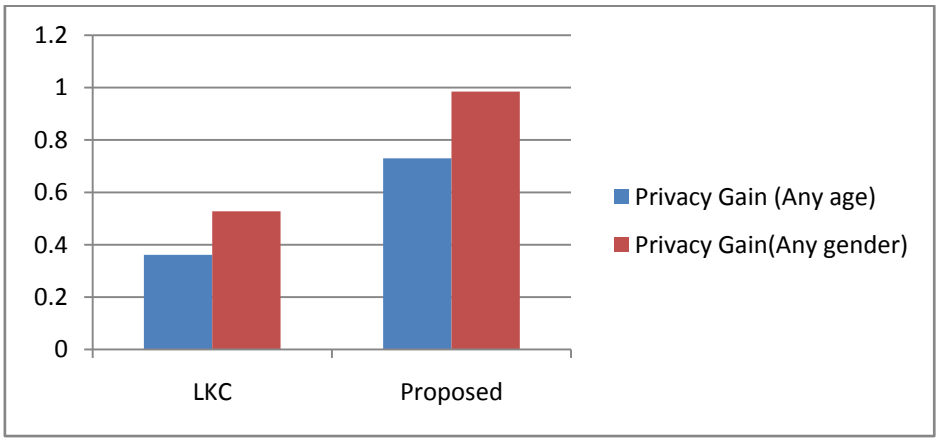
Performance of the Proposed method on Anonymized Dataset

Methods	*TDR(Dataset 1)*	*Proposed Method*
Privacy Gain Any Age	0.3958	0.493
Privacy Gain Any Gender	0.4280	0.985

Privacy gain of the proposed method compared to TDR Model



Privacy gain of the proposed method compared to LKC Model



Summary:
The outcome of this research project are the design of three different methods to transform the published dataset into anonymyzed dataset which preserves the privacy of data owners. They were tested with published datasets in government and health websites.